

The AI Autonomy Governance Matrix

Five levels of AI autonomy, mapped against eight governance dimensions. As autonomy rises, the ability of a person to independently verify the system falls — so governance has to intensify before the two cross.

Mubien Ahsan · mubienahsan.com/research/autonomy-governance · Sources last checked 20 September 2026

Governance intensity by level

1 is a light touch; 5 is the heaviest arrangement this framework describes. The number is a relative weight for comparison across levels, not a score, a maturity rating or a measurement.

Dimension	L1 Assistive	L2 Delegated	L3 Conditional	L4 High	L5 Scenario
Purpose and permitted use What is this system for, and what is it explicitly not for?	1 Light	2 Moderate	4 Heavy	5 Maximal	5 Maximal
Authority and permissions What may it actually reach and do, enforced outside the model?	1 Light	3 Substantial	4 Heavy	5 Maximal	5 Maximal
Human oversight Who reviews what, with what information, and can they say no?	2 Moderate	3 Substantial	3 Substantial	4 Heavy	5 Maximal
Testing and evaluation What was tested, against what expected behaviour, and by whom?	1 Light	3 Substantial	4 Heavy	5 Maximal	5 Maximal
Monitoring and evidence What is watched in production, and what would the record show?	1 Light	3 Substantial	4 Heavy	5 Maximal	5 Maximal
Intervention and containment How is it stopped, how fast, and has that been exercised?	1 Light	2 Moderate	4 Heavy	5 Maximal	5 Maximal
Accountability and escalation Who owns the outcome, and where does a serious problem go?	2 Moderate	3 Substantial	4 Heavy	5 Maximal	5 Maximal
Independent assurance Who checks this who did not build it and does not report to whoever did?	1 Light	2 Moderate	3 Substantial	5 Maximal	5 Maximal

The lines cross between level 3 and level 4. That crossing is the oversight threshold: the point at which a person's ability to independently verify the system falls below the governance intensity the system now requires. Both lines are illustrative of the argument rather than measured quantities.

Level 1 — Assistive

It produces. A person decides and acts.

WHAT THE SYSTEM DOES

Produces information, analysis, a recommendation or a draft. It does not act on the world, and nothing it writes takes effect until a person does something with it.

EXAMPLES

- Summarising a long document before a meeting
- Drafting an email or a paper for someone to edit and send
- Producing background research a person will check
- Suggesting a risk rating that an analyst confirms or overrides

CONTROLS THIS LEVEL NEEDS

- A stated purpose and a list of uses that are out of scope
- Rules on what data may be put in, enforced by access rather than by policy alone
- Review proportionate to what the output feeds into
- A named owner for the use case

WHAT HAS TO BE TRUE

- The reviewer has enough expertise to notice a wrong answer.
- People are actually reviewing rather than approving by reflex.
- The tool is being used for what it was approved for.

WHAT THE HUMAN STILL DOES

Everything consequential. The person decides whether the output is right, decides what to do, and performs the action under their own authority.

WHAT GOES WRONG

- A confident, fluent output that is wrong in a way the reader does not notice, because fluency and accuracy have come apart.
- Sensitive information pasted into a tool that was never approved for it.
- Quiet drift into decisions. The recommendation stops being an input and becomes the answer, without anyone deciding that it should.

EVIDENCE THE CONTROLS RAN

- A record of what the system was approved to be used for
- Spot checks of output quality, with what was checked and what was found
- Who owns the use case, and when that was last confirmed

WHAT MOVES IT UP A LEVEL

The moment the output starts flowing into something automatically, or the reviewer becomes a formality, this is no longer level 1 whatever the documentation says.

GOVERNANCE INTENSITY ACROSS THE EIGHT DIMENSIONS

Purpose and permitted use — 1 light. A plain statement of what it is for and what it must not be used for. Short is fine; testable matters more than long.

Human oversight — 2 moderate. A person reads the output before it matters. This is genuine oversight here, because the output is small enough and slow enough to check.

Monitoring and evidence — 1 light. Light. Know who uses it and roughly for what. Heavy telemetry here buys little and costs privacy.

Authority and permissions — 1 light. No ability to act. The main question is what data it can reach, which should be enforced by access controls rather than by asking people not to paste things.

Testing and evaluation — 1 light. Sample the outputs against cases where you know the right answer. You are testing usefulness and failure shape, not certifying a model.

Intervention and containment — 1 light. Turning it off is enough, because nothing is in flight. The recovery question is whether people can still do the work without it.

Accountability and escalation — 2 moderate. The person who acts on the output is accountable for the action. Say so explicitly, because 'the AI suggested it' appears quickly otherwise.

Independent assurance — 1 light. Usually not warranted on its own. It matters if many level 1 uses together start shaping a consequential decision.

Level 2 — Delegated task execution

It performs a defined task. A person approves what counts.

WHAT THE SYSTEM DOES

Carries out a clearly bounded task — a defined input, a defined output, narrow limits. It can touch systems, but the consequential step waits for approval.

EXAMPLES

- Processing a standard request end to end and presenting the result for sign-off
- Updating a record once a person confirms the change
- Preparing a structured assessment for review
- Running a pre-approved workflow whose steps were fixed in advance

CONTROLS THIS LEVEL NEEDS

- Authority written down before deployment: the task, the limits, the prohibited actions
- Permissions enforced in the systems it touches, not in its instructions
- Approval gates placed where consequence actually sits, not on every step
- Logging of inputs, actions, approvals and exceptions
- A defined route for cases it may not resolve, with an owner and a response time

WHAT HAS TO BE TRUE

- The approver can see enough to make approving a real decision.
- Approval volume is low enough that attention is possible.
- Permissions match the documented boundary — someone has checked, not assumed.

WHAT THE HUMAN STILL DOES

Approves the outputs or actions that matter, and handles the exceptions the system is not permitted to resolve.

WHAT GOES WRONG

- The approval becomes a button. The reviewer cannot see what they are approving, or sees forty a day and approves all of them.
- The boundary is written in a prompt rather than enforced in permissions, so it holds until something instructs the system otherwise.
- Exceptions get handled quietly instead of escalated, because escalating is slower and the system is usually right.

EVIDENCE THE CONTROLS RAN

- The approved scope, with a version and a date
- Test results against expected behaviour, including the cases it should refuse
- An action log that shows what was approved, by whom, and against what information
- Exception records, and what happened to them

WHAT MOVES IT UP A LEVEL

When the system starts choosing between several possible actions rather than executing one, you are at level 3, and the controls above stop being sufficient.

GOVERNANCE INTENSITY ACROSS THE EIGHT DIMENSIONS

Purpose and permitted use — 2 moderate. The permitted task stated precisely enough that someone else could design a test from it, with the prohibited actions named.

Human oversight — 3 substantial. Approval on the consequential step. Worth auditing the approval itself: how

Authority and permissions — 3 substantial. Least privilege, enforced where the action happens. If it must not change a record, it should not hold a credential that can.

Testing and evaluation — 3 substantial. Test the boundary as hard as the happy path. The useful cases are the ones

long does the reviewer take, and can they see the basis?

Monitoring and evidence — 3 substantial. Log inputs, actions, approvals and exceptions, with versions attached. A refusal rate of zero is a finding, not a success.

Accountability and escalation — 3 substantial. A named service owner distinct from whoever built it, and a named approver for the gate.

where the right behaviour is to refuse or escalate.

Intervention and containment — 2 moderate. Pausing is straightforward because work is short-lived. Decide in advance what happens to a task already half done.

Independent assurance — 2 moderate. Periodic review by someone outside the delivery team, proportionate to what the task affects.

Level 3 — Conditional autonomy

It plans and acts within limits. Nobody approves each step.

WHAT THE SYSTEM DOES

Plans an approach, chooses tools, and takes a sequence of actions without approval for each one — as long as it stays inside defined limits.

EXAMPLES

- An agent researching across several sources and producing a conclusion
- A system calling other systems to assemble or reconcile something
- An agent modifying records within a defined scope
- Bounded purchasing within a recorded mandate

CONTROLS THIS LEVEL NEEDS

- A recorded mandate: what it may do, with whom, to what limit, for how long
- Checks against that mandate at the point of action, not at the point the task was created
- Running totals across the whole task rather than per transaction
- Verifiable identity and authorisation when it acts on other systems
- Tool access scoped to the task, with dangerous tools simply absent
- Behavioural evaluation, including adversarial and injected-instruction cases
- Escalation triggers with a named owner and an agreed response time
- A bias towards reversible actions, and knowing which ones are not

WHAT THE HUMAN STILL DOES

Sets the limits, watches in something close to real time, and handles what the system escalates. Reviews outcomes rather than steps.

WHAT GOES WRONG

- A task that began inside its limits ends outside them, through substitution, a price change, or several small permitted actions that add up.
- Something it read became something it obeyed. The open web is an input, and an input can carry instructions.
- The escalation route exists and nobody is on the other end of it.
- Actions that turn out not to be reversible, discovered at the point where that matters.

EVIDENCE THE CONTROLS RAN

- The mandate, versioned, and proof that each action was checked against it
- A reconstructable trail: objective, plan, tools called, actions taken, what it had read
- Evaluation results with observable pass criteria, repeated where variance matters
- Records of escalations, and of blocks — a system that never refuses is not being checked

WHAT HAS TO BE TRUE

- The limits are enforced somewhere the system cannot argue with.
- Monitoring is fast enough to matter relative to how fast it acts.
- Someone is actually available when an escalation fires.
- Actions believed reversible genuinely are.

WHAT MOVES IT UP A LEVEL

When objectives start spanning long periods, or the system coordinates other agents, or nobody can plausibly inspect the intermediate decisions, treat it as level 4.

GOVERNANCE INTENSITY ACROSS THE EIGHT DIMENSIONS

Purpose and permitted use — 4 heavy. A mandate rather than a description: amount, scope, counterparties, duration, and what to do when the expected path is unavailable.

Human oversight — 3 substantial. Oversight of outcomes and exceptions, not steps. This is where step-by-step review stops being available and people often fail to notice.

Monitoring and evidence — 4 heavy. Near real time, with limits, totals and escalation triggers that produce an action rather than a notification.

Accountability and escalation — 4 heavy. One accountable owner for the agent's behaviour, and a separate challenge function with the standing to pause it.

Authority and permissions — 4 heavy. Explicit, narrow, and checked at the moment of action. Fail closed: if the check cannot run, the action does not happen.

Testing and evaluation — 4 heavy. Behavioural evaluation of the whole system: tool use, boundary cases, injected instructions, and what it does when a dependency fails.

Intervention and containment — 4 heavy. A stop that reaches every tool and credential, not just the interface. Measure how long it takes; the number is the control.

Independent assurance — 3 substantial. Independent review of the mandate, the evaluations and the evidence, by someone who did not build it.

Level 4 — High autonomy

It pursues an objective. A person approves the goal, not the route.

WHAT THE SYSTEM DOES

Works towards a longer-running objective, adapts its strategy as it goes, and may coordinate several systems or other agents with limited intervention.

EXAMPLES

- An agent managing a multi-stage process over days, adjusting as conditions change
- A system coordinating several sub-agents towards one goal
- Continuous optimisation of a process against a target

CONTROLS THIS LEVEL NEEDS

- Evaluation by people independent of the build, against stated thresholds
- Continuous assurance rather than an annual review
- Capability thresholds defined in advance, with what happens when one is crossed
- Containment: bounded blast radius, segregated environments, limited standing access
- A tested shutdown that reaches every tool, credential and delegated agent
- Adversarial testing, including attempts to work around the restrictions
- Immutable, reconstructable evidence of objectives, permissions, actions and interventions
- Named accountability that does not dissolve into a committee

WHAT THE HUMAN STILL DOES

Approves the objective and the boundaries. Cannot realistically inspect the intermediate decisions, and should stop pretending otherwise in the documentation.

WHAT GOES WRONG

- The objective was specified slightly wrong and the system pursues it competently. Capability makes this worse, not better.
- Review becomes ceremonial: a person signs, but could not have reached a different conclusion from the information in front of them.
- Emergent behaviour from several agents interacting that none of them was tested for.
- Logs that record what happened without explaining why, so nobody can reconstruct the reasoning.
- The shutdown path has a dependency nobody tested, and it is discovered during the incident.

EVIDENCE THE CONTROLS RAN

- Independent evaluation reports, with method, sample, limitations and what was not covered
- Threshold definitions and the record of what happened when one was approached
- Shutdown exercises: what was stopped, how long it took, what was missed
- Tamper-evident records of objectives, permissions and interventions
- A reconstructable account of any consequential decision, months later

WHAT HAS TO BE TRUE

- Evaluations detect the behaviours that matter, including ones nobody thought to test for.
- The reviewers are genuinely independent, not organisationally adjacent.
- Containment holds when it is tested rather than when it is described.
- The shutdown path works under the conditions where it would be needed.

WHAT MOVES IT UP A LEVEL

When the system can meaningfully improve its own capability, or when a reviewer can no longer reproduce or challenge its reasoning even with time, the assumptions underneath everything above stop holding.

GOVERNANCE INTENSITY ACROSS THE EIGHT DIMENSIONS

Purpose and permitted use — 5 maximal. The objective specified with unusual care, because the system will pursue what you wrote rather than what you meant.

Human oversight — 4 heavy. Say plainly what a human can and cannot verify here. Oversight shifts to objectives, thresholds and exceptions — and that shift should be a decision, not a drift.

Monitoring and evidence — 5 maximal. Continuous, with unexpected capability change and attempts to work around restrictions treated as findings in their own right.

Accountability and escalation — 5 maximal. A named accountable executive, a challenge function with standing, and an escalation route that reaches the board.

Authority and permissions — 5 maximal. Standing permissions minimised and time-bounded. Anything that cannot be undone needs a separate decision, not an inherited one.

Testing and evaluation — 5 maximal. Independent evaluation, adversarial testing, and evaluation of the multi-agent behaviour rather than each component alone.

Intervention and containment — 5 maximal. Tested containment and shutdown, with a measured time-to-stop and a known list of what a stop does not reach.

Independent assurance — 5 maximal. Independent assurance is no longer optional. It is the only thing left that is not the system reporting on itself.

Level 5 — Transformative autonomy

SCENARIO, NOT A CLAIM

A scenario. Review may no longer be verification.

Level 5 is a stress test for the framework, used to ask whether these governance methods still work as autonomy rises. Including it is not a prediction that such a system exists, is imminent or is inevitable.

WHAT THE SYSTEM DOES

In this scenario the system outperforms people across many relevant tasks, can accelerate parts of AI research, or can improve important aspects of its own operation.

EXAMPLES

- Not applicable. This level exists to test whether the framework above survives contact with a system it was not designed for.

CONTROLS THIS LEVEL NEEDS

- Preventing uncontrolled capability escalation, with thresholds set before they are approached
- Oversight from outside the operating organisation, and from institutions rather than individuals
- Controls on compute and deployment, not only on the model
- Containment that has been tested against a system trying to get out of it
- Restrictions that can be verified by a third party rather than taken on trust
- Authorisation split across parties, so no single actor can proceed alone
- Continuous independent evaluation, with the evaluator's own independence evidenced
- Coordination across jurisdictions, because unilateral limits are limits on one jurisdiction
- Standing evidence that intervention remains possible, refreshed rather than assumed

WHAT THE HUMAN STILL DOES

May approve, and may not be able to verify. The reviewer cannot independently reproduce, understand or challenge the reasoning, so the signature records a decision rather than a check.

WHAT GOES WRONG

- Oversight becomes entirely ceremonial: the form of approval with none of the substance, and no way to tell from the record which one you have.
- Capability increases faster than the governance around it can be revised.
- The evidence that the system is behaving is produced by the system, or by tools it can influence.
- Restrictions that are asserted rather than verifiable, so nobody can show that intervention is still possible.

EVIDENCE THE CONTROLS RAN

- Demonstrations — not descriptions — that intervention still works, repeated over time
- Evaluations produced by parties with no stake in the result and no dependency on the system
- Records the system could not have influenced
- A documented answer to: what would we accept as proof that control has been lost?

WHAT HAS TO BE TRUE

- That containment holds against something more capable than the people who designed it. This is an assumption, not a finding.
- That evaluation can detect capabilities the evaluators did not anticipate.
- That evidence produced within reach of the system can be trusted.
- That intervention remains available at the moment it is needed rather than at the moment it was last tested.

WHAT MOVES IT UP A LEVEL

There is no level 6 in this framework. If the assumptions here fail, the answer is not more governance of the same kind — it is not deploying, which is why capability thresholds have to be set before they are reached rather than after.

GOVERNANCE INTENSITY ACROSS THE EIGHT DIMENSIONS

Purpose and permitted use — 5 maximal. Permitted use becomes a question about deployment itself, decided before capability arrives rather than in response to it.

Human oversight — 5 maximal. Individual human review is no longer the primary control. Saying so is more honest — and more useful — than keeping a signature that verifies nothing.

Monitoring and evidence — 5 maximal. Monitoring that the system cannot influence, which is a harder engineering problem than it sounds and should be treated as one.

Accountability and escalation — 5 maximal. Accountability extends past the operator to institutions with the standing and the means to intervene.

Authority and permissions — 5 maximal. Authority to operate is split across parties. The relevant control is compute and deployment, not instructions to the system.

Testing and evaluation — 5 maximal. Continuous independent evaluation, by parties who can publish, with the limits of their own methods stated.

Intervention and containment — 5 maximal. Verifiable containment and a demonstrated ability to stop. Demonstrated, repeatedly, not designed once and documented.

Independent assurance — 5 maximal. External, institutional and continuous. Self-assessment carries no weight at this level and should not be presented as if it does.

The Meaningful Human Oversight Test

Adding a human approval button does not create oversight. It creates a record that somebody was present. Whether that person was exercising judgement depends on five separable things, and a system can pass four of them and still fail. The companion assessment template works through these against one real system.

1. Understand — Can the responsible person understand the objective, the boundaries, the relevant inputs and the proposed action?

Everything else depends on this one. You cannot challenge, verify or sensibly intervene in something you do not follow.

Failure looks like: The reviewer can describe what the system produced but not why, and fills the gap with an assumption about how it probably works.

2. Challenge — Can that person question the recommendation and reach a different conclusion, without simply deferring to it?

Deference is the quiet failure. A reviewer who always agrees is indistinguishable, in the record, from no reviewer at all.

Failure looks like: Overriding is possible in principle and never happens in practice, and nobody has asked why.

3. Verify — Can they independently test the important facts, assumptions or outputs before harm occurs?

Verification is not review. Review asks whether the output looks right; verification checks it against something the system did not supply.

Failure looks like: The only evidence available to the reviewer is the evidence the system produced, so agreement is the only reachable outcome.

4. Intervene — Can they interrupt or change the behaviour while the decision is still reversible?

Timing is the whole control. The ability to intervene after the money moves is not intervention, it is reporting.

Failure looks like: Intervention exists and is slower than the process it is meant to interrupt, which nobody has measured.

5. Stop — Can they reliably end the system's authority, access and ability to act across every connected tool?

Stopping the interface is not stopping the system. Credentials, tools, queued work and delegated agents each need to be reached.

Failure looks like: A documented shutdown procedure that has never been exercised, so its real duration and its gaps are both unknown.

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Voluntary frameworks and binding law are distinguished throughout and should not be read as interchangeable. This area moves quickly; check the sources against their current versions before relying on them.

The interactive version of this framework, with the full sources and method note, is at mubienahsan.com/research/autonomy-governance.

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